Colour Light Signals

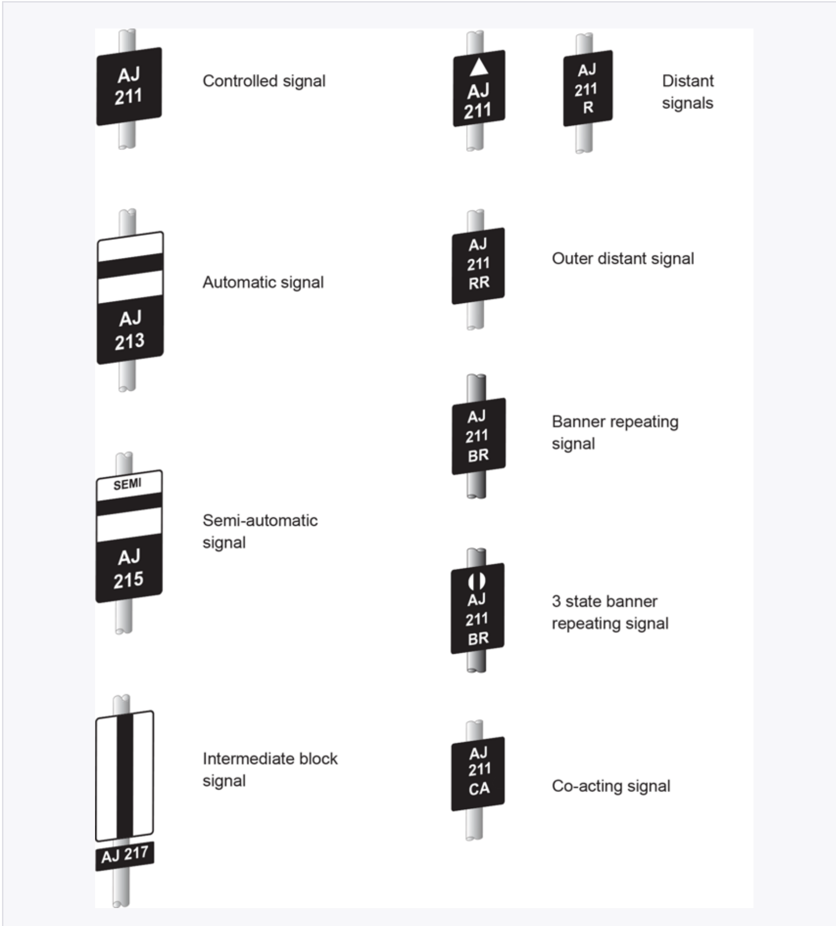
Revision Notes

Sources: RS521 Issue 9 | GERT8000-S7 Issue 7

1. Signal Identification Plates

Every signal has an identification plate attached to its post. The plate tells you what type of signal it is. You must know the plate types by sight.

Plate appearance	Signal type	What it means
Plain black plate, white text (e.g. AJ 211)	Controlled signal	Operated by the signaller
Black plate with white horizontal stripe across top	Automatic signal	Operated by the passage of trains. Signaller can place at danger using signal post replacement switch.
"SEMI" above the plate	Semi-automatic signal	Normally operated by passage of trains but can also be controlled from a signal box or ground frame.
Black plate with white vertical stripe	Intermediate block signal	A stop signal controlling exit from an intermediate block section into an absolute block section.
Upward arrow + letter "R"	Distant signal	Cannot show a stop aspect. Warns of stop signals ahead.
Letters "RR"	Outer distant signal	A second distant signal further in advance of the first distant.
Letters "BR"	Banner repeating signal	Repeats the aspect of a signal with restricted sighting.
Letters "CA"	Co-acting signal	Mirrors the main signal to aid sighting from close range.



Every identification plate type shown together — RS521 §1.2

KEY RULE (RS521 §1.2): Some colour light distant signals are identified by a white triangle OR the letters "R" or "RR" on the signal identification plate.

2. How Colour Light Signalling Works

2.1 Track Circuits

Track circuits detect whether a section of line is occupied. You need to understand the four main components:

Component	Function
Power source	Supplies electrical current through the rails of the track circuit section.
Insulated rail joints (insulated gaps)	Electrically isolate one track circuit section from the next, creating defined boundaries.
Track circuit relay	Energised (picked up) when the section is clear. De-energised (dropped) when a train short-circuits the rails — which holds the signal at danger.
Track circuit (the rails)	Carries the current between the power source and the relay. A train's axles short-circuit the current, dropping the relay.

Axle counters are an alternative to track circuits. They count axles as a train enters and exits a section to confirm the complete train has cleared.

2.2 Overlap and Clearance Point

These are protection distances beyond a signal:

Term	Distance	Applies to
Overlap	200 metres beyond the signal	Colour light signalling — the area kept clear beyond a stop signal in case a train overruns it.
Clearance point	400 metres beyond the signal	Absolute block signalling — the distance a train must have passed before the section ahead is considered clear.

REMEMBER: Overlap = 200m (colour light). Clearance point = 400m (absolute block).

3. Signal Aspect Sequences

REMEMBER: Red is ALWAYS at the BOTTOM of a colour light signal. Always reduce speed at a caution signal.

3.1 Two-Aspect Signalling

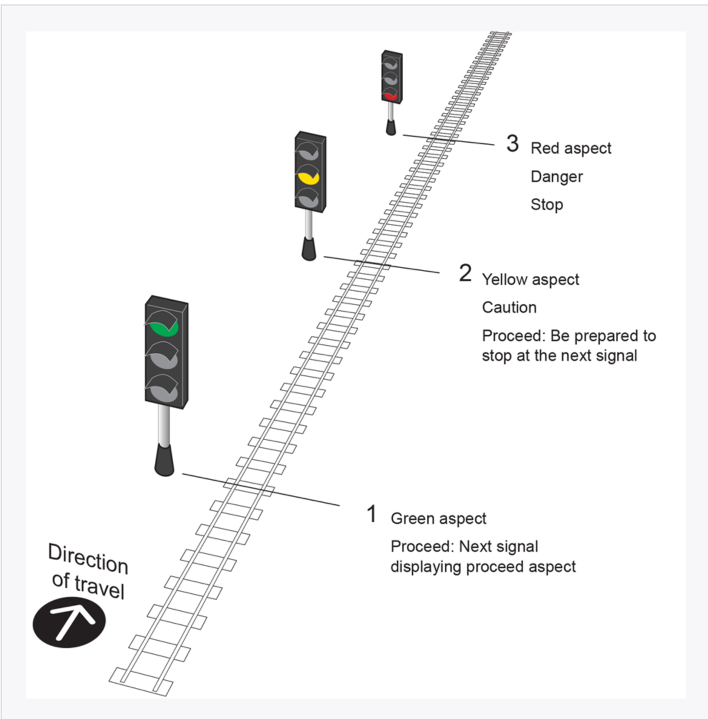
Two-aspect signalling uses a distant signal followed by a stop signal. There is no multi-aspect sequencing — you get one warning before the stop.

Aspect	Colour	Meaning
Distant at caution	Yellow	Be prepared to stop at the next stop signal.
Distant clear	Green	The associated stop signal is clear — proceed.
Stop at danger	Red	Stop. Do not pass.
Stop clear	Green	Proceed.

3.2 Three-Aspect Signalling — Normal Sequence

Three-aspect signalling uses Red, Yellow, and Green. There is no separate distant signal — the sequence of aspects provides the warning.

Signal ahead shows	This signal shows	Meaning to driver
Red (danger)	Single Yellow	Caution — be prepared to stop at the NEXT signal.
Single Yellow (caution)	Green	Proceed — line is clear.
Green (clear)	Green	Proceed — line is clear.



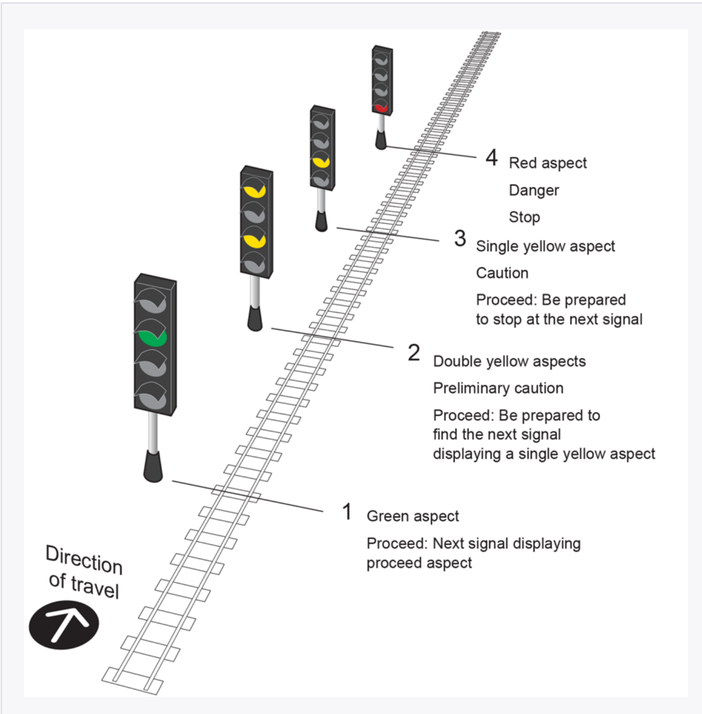
Three-aspect signalling — the normal sequence — RS521 §2.1

NOTE (GERT8000-S7 §1.6): If you stop or nearly stop at a 2-aspect colour light signal at danger (one that cannot display a yellow aspect) and it then clears, be prepared to stop at the next stop signal worked by the same signalbox.

3.3 Four-Aspect Signalling — Normal Sequence

Four-aspect signalling adds Double Yellow, giving two stages of advance warning.

Signal ahead shows	This signal shows	Meaning to driver
Red (danger)	Single Yellow	Caution — be prepared to stop at the NEXT signal.
Single Yellow	Double Yellow	Preliminary caution — be prepared to find the next signal at single yellow.
Double Yellow or Green	Green	Proceed — line is clear.



Four-aspect signalling — the normal sequence — RS521 §2.2

Key rule from GERT8000-S7 §1.5 — Signal not showing correctly:

Situation	Treat as
Stop signal not showing / aspect not clear or obvious / white light showing	Signal at DANGER
Distant signal not showing correctly	Signal at CAUTION
Position-light, subsidiary or shunting signal not showing correctly	At NORMAL (stop) indication

4. Junction Indicators and Route Indicators

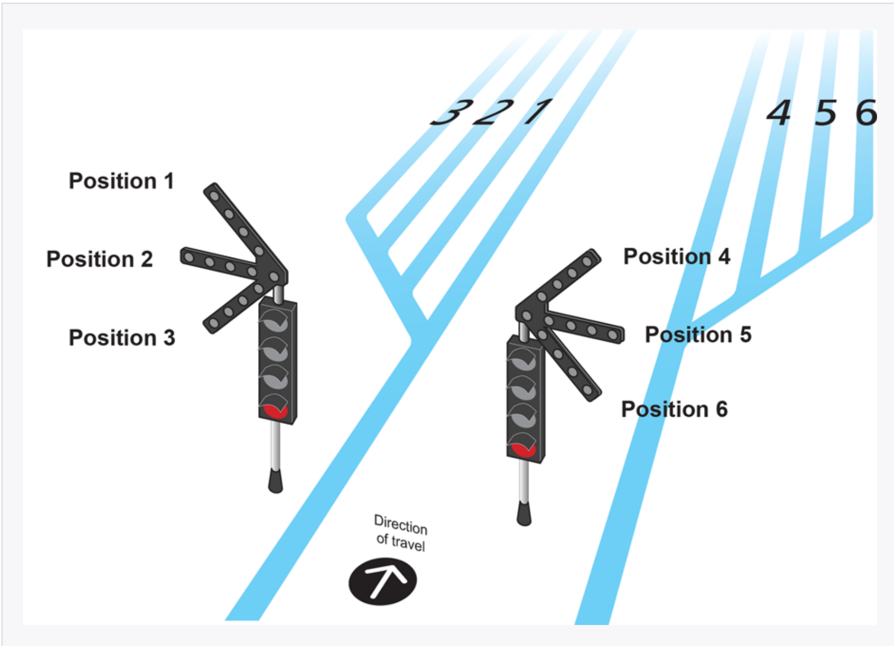
4.1 Junction Indicators

Junction indicators show that a train is being signalled to a route to the left or right of the straight route. They display a line of white lights when a proceed aspect is shown, normally above the signal.

- When the straight route is obvious, there is normally no junction indicator for that route.
- Where there is no obvious straight route, a junction indicator is provided for all signalled routes.
- Where the straight route is not the highest-speed route, the junction indicator normally applies to the lower-speed route.
- Where diverging routes ahead are both equal speed, a junction indicator is provided for each route.

Reading junction indicators:

- Feathers to the LEFT of the signal = routes 1, 2, 3 (furthest left = route 1).
- Feathers to the RIGHT of the signal = routes 4, 5, 6 (furthest right = route 6).
- Wait until the main aspect clears before proceeding — do not act on the junction indicator alone.



Junction indicator feather positions, both sides of the signal — RS521 §2.3

IMPORTANT: When waiting for a route indicator, once it lights up, wait for the main aspect before proceeding — never act on the indicator alone.

4.2 Route Indicators

At some locations a route indicator showing a letter or number is provided instead of, or in addition to, a junction indicator. The letter or number shows the route onto which the movement is being signalled.

A route indicator showing a numbered route — RS521 §2.4

Route indicators associated with position-light signals are of miniature design.

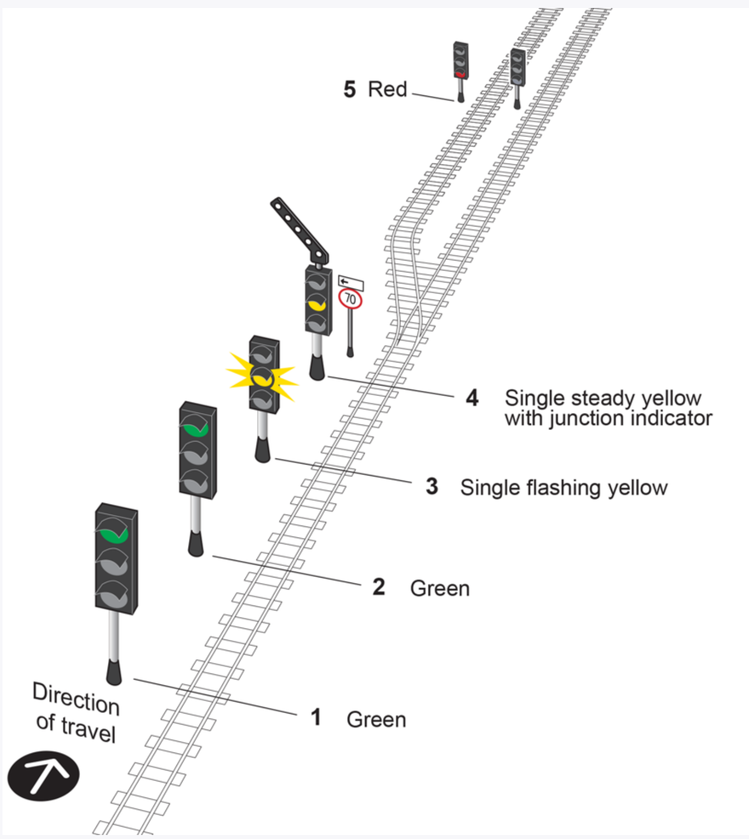
5. Flashing Yellow Aspects

A flashing yellow aspect means facing points at a junction ahead are set for a diverging route with a lower speed than that for the straight route.

KEY POINT: A flashing yellow confirms the route ahead is set for a diverging (lower-speed) route at the junction — advance information layered on top of the normal caution meaning.

5.1 Three-Aspect Flashing Yellow Sequence

Signal	Aspect shown
Signal 1	Green
Signal 2	Green
Signal 3	Single flashing yellow
Signal 4 (junction signal)	Single steady yellow + junction indicator
Signal 5	Red

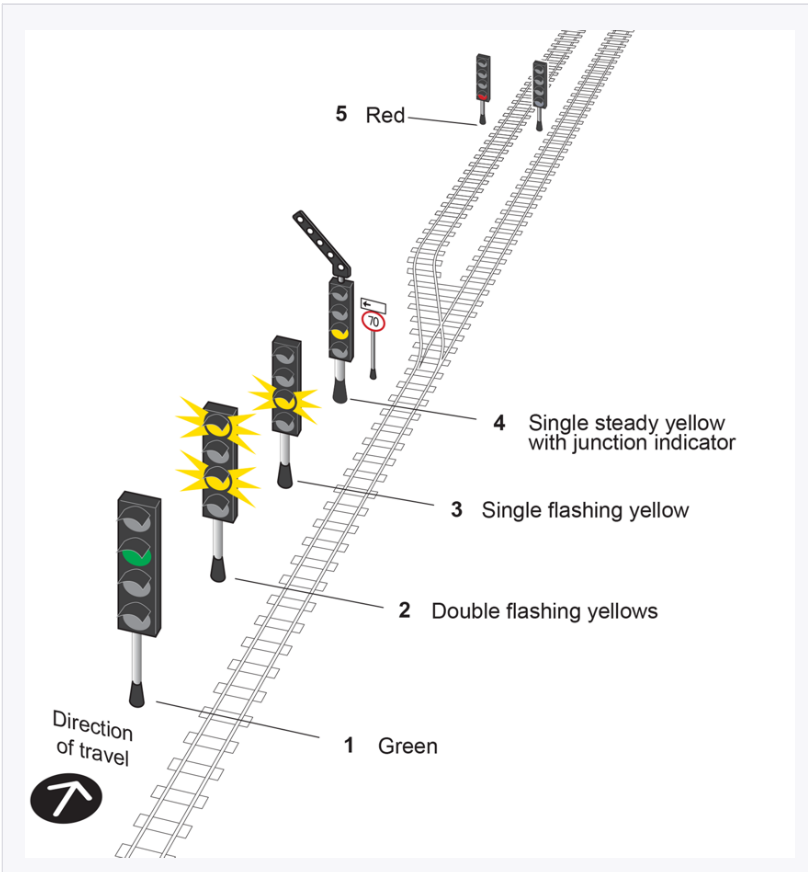


Three-aspect flashing yellow signalling — the normal sequence — RS521 §2.5

When a single steady yellow aspect is displayed together with a junction indicator at signal 4, this has the normal meaning of a yellow aspect: be prepared to stop at the next signal (5). This applies even though a flashing aspect may have been displayed at signal 3.

5.2 Four-Aspect Flashing Yellow Sequence

Signal	Aspect shown
Signal 1	Green
Signal 2	Double flashing yellows
Signal 3	Single flashing yellow
Signal 4 (junction signal)	Single steady yellow + junction indicator
Signal 5	Red



Four-aspect flashing yellow signalling — the normal sequence — RS521 §2.5

If the train is between signals 2 and 3 when signal 4 is cleared for the diverging route, signal 3 may then display one flashing yellow aspect. This applies even though a steady aspect has been displayed at signal 2.

When a single steady yellow aspect is displayed together with a junction indicator at signal 4, this has the normal meaning of a single yellow aspect: be prepared to stop at the next signal (5). This applies even though a flashing aspect may have been displayed at signal 3.

5.3 Preliminary Caution Route Indicator

A preliminary route indicator gives advance information about the route set beyond a junction signal ahead. It mimics the junction indicator — displaying an arrow pointing in the same direction.

- 4-aspect signalling areas: 3 preliminary route indicators are provided.

- 3-aspect signalling areas: 2 preliminary route indicators are provided.
- If the junction signal is at danger, the preliminary route indicator is not illuminated.
- If the junction signal is cleared without a junction indicator (straight route), the preliminary route indicator shows an arrow pointing straight up.

5.4 Risks with Flashing Yellow Signalling

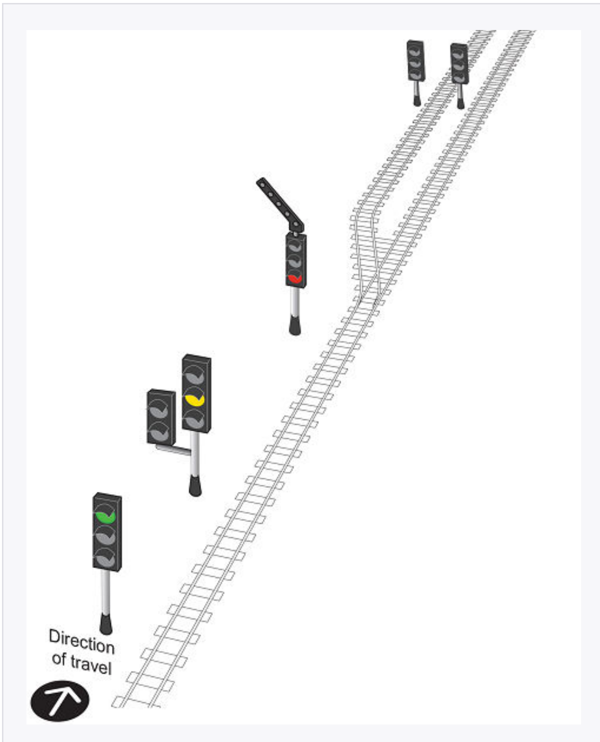
Travelling between different signalling systems creates risk. Key risks include:

- Entering a flashing yellow area from a non-flashing area — driver may not anticipate the signal sequence changing.
- Returning to a steady-aspect area — the yellow reverts to normal caution meaning; be prepared to stop at the next signal.
- For trains on which ERTMS is operating: the ability for signals to display flashing aspects is DISABLED. Only standard aspect sequences are shown to these trains. Route/junction indicators continue to operate.

6. Splitting Distant Signals

Splitting distant signals are provided at diverging junctions to show which route is set. They may be used with three or four-aspect signalling.

- The primary head relates to the main route.
- The off-set head (displaced to left or right) relates to the diverging route.
- When a junction signal is at danger, no aspect is shown in the head(s) for that route.
- When the junction is cleared for the straight route, the primary heads show the aspect beyond; the off-set heads show aspects appropriate for danger on the diverging route.
- When the junction is cleared for the diverging route, the off-set heads show the aspect beyond; the primary heads show aspects appropriate for danger on the straight route.



Splitting distant signals approaching a junction, with the diverging route's off-set heads — RS521 §2.6

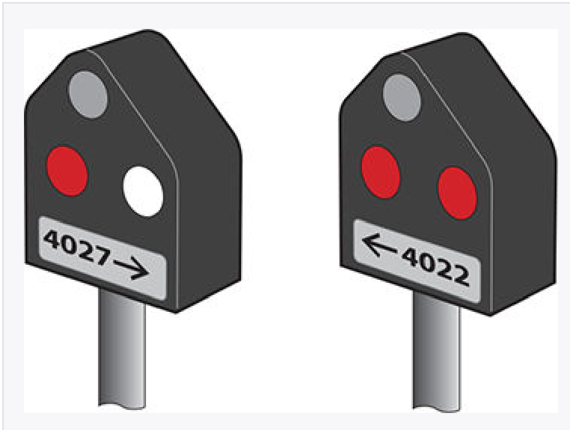
7. Position-Light Signals

Position-light signals are used at ground level to authorise movements — typically for shunting, entering sidings, or permissive working.

7.1 Red Aspect Position-Light Signals

These are stop signals. They are normally positioned at ground level independently of a main aspect.

Indication	Meaning
Two lights in a line (horizontal) — red	STOP
Two lights at 45 degrees — white	Proceed at caution towards the next train, signal or buffer stop — be prepared to stop short of any obstruction.



Red aspect position-light signals — RS521 §2.7

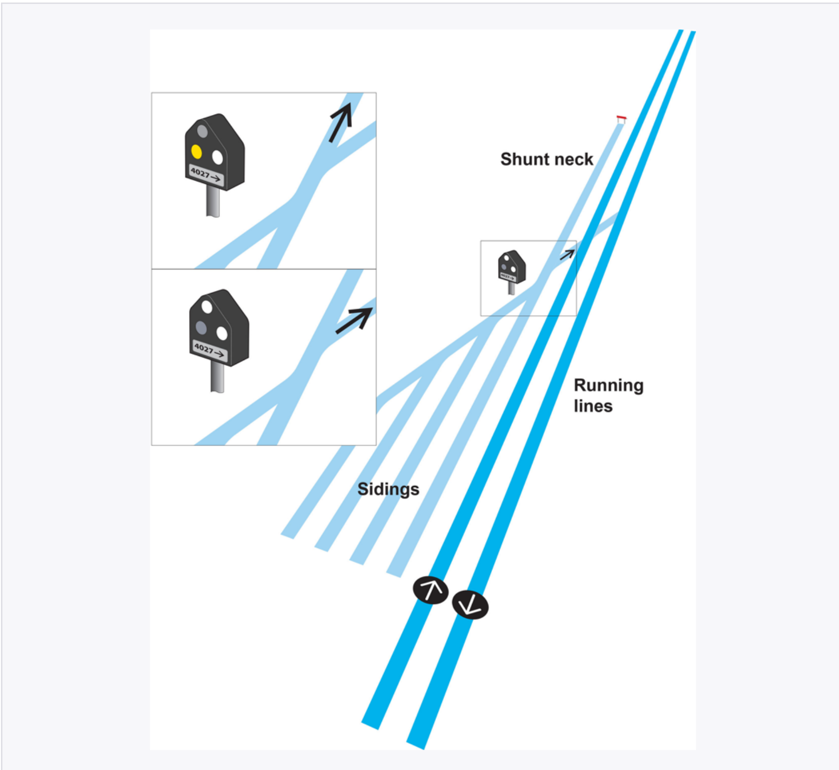
When proceeding on the authority of a main aspect, any position-light signals along the route between main running signals will show a proceed indication.

7.2 Yellow Aspect Position-Light Signals (Shunting)

These apply only to movements in the direction to which the signal can be cleared. Other movements can pass the signal without it being cleared.

Indication	Meaning
Two lights in a line (horizontal) — yellow	STOP. May be passed if the movement is heading to a shunt neck or siding, not the running line. The route may be obstructed — including by a train.
Two lights at 45 degrees — white	Proceed at caution as far as the line is clear.

Yellow aspect position-light signals — RS521 §2.7

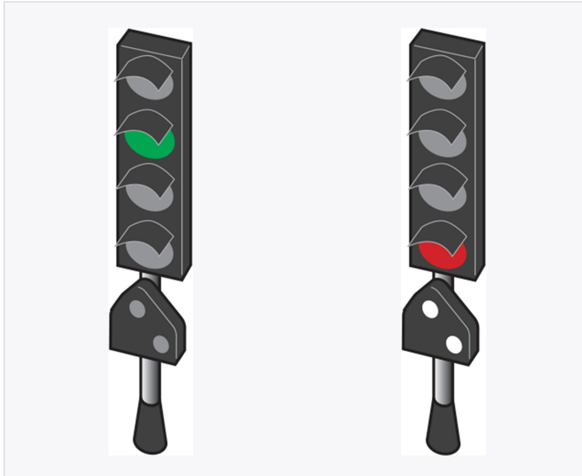


How a yellow position-light signal is used approaching a shunt neck and sidings — RS521 §2.7

7.3 Position-Light Signals Associated with a Main Aspect

These are normally positioned below the main aspect on the same signal post.

- Normal state: UNLIT — meaning obey the main signal.
- When showing two white lights at 45°: proceed at caution towards the next train, signal or buffer stop — be prepared to stop short of any obstruction.



Position-light signal associated with a main aspect — obey the main signal (green, left) — RS521 §2.7

GERT8000-S7 §3.1 — CRITICAL RULE: A driver must NOT proceed with a PASSENGER TRAIN on the authority of a position-light signal or semaphore shunting signal, UNLESS entering a permissive platform line on the authority of a position-light or subsidiary signal associated with a main aspect.

GERT8000-S7 §3.2: If a position-light or subsidiary signal is cleared but the normal route indication is not shown — move at caution, be prepared to stop before any obstruction.

7.4 Permissive Working with a Position-Light Signal

When a proceed indication is given on a position-light signal with the main aspect at red:

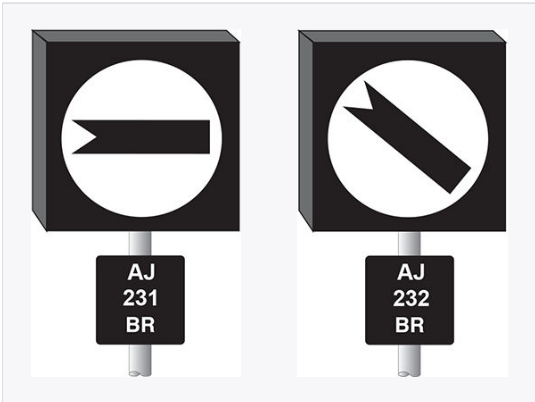
1. Always drive at caution.
2. Be prepared to stop short of any obstruction, including a train, vehicle, or buffer stop.
3. The route may be occupied by another train or vehicle ahead.
4. Do not assume the line is clear.

8. Banner Repeating and Co-Acting Signals

8.1 Banner Repeating Signals

Banner repeating signals are provided on the approach to signals with restricted sighting (e.g. curves, tunnels, buildings) to give advance warning of the signal aspect.

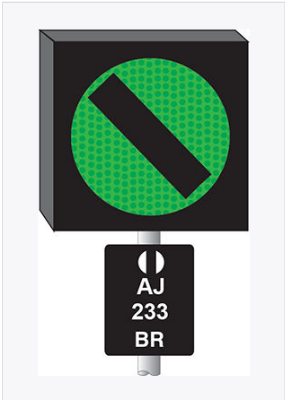
Banner repeater position	Meaning
ON (horizontal arm or light)	The signal to which it applies is at danger (or a distant is at caution).
OFF (arm or light at 45°)	The signal to which it applies is displaying a proceed aspect.



Banner repeater ON (left, signal at danger) and OFF (right, signal clear) — RS521 §5.7

8.2 Two-State vs Three-State Banner Repeaters

Type	States	Additional meaning
2-state	ON / OFF	OFF = signal is showing a yellow OR green (proceed) aspect.
3-state	ON / OFF (yellow) / OFF (green)	3-state adds a green indication: if the banner shows a green-lit OFF, the signal ahead is showing green specifically.

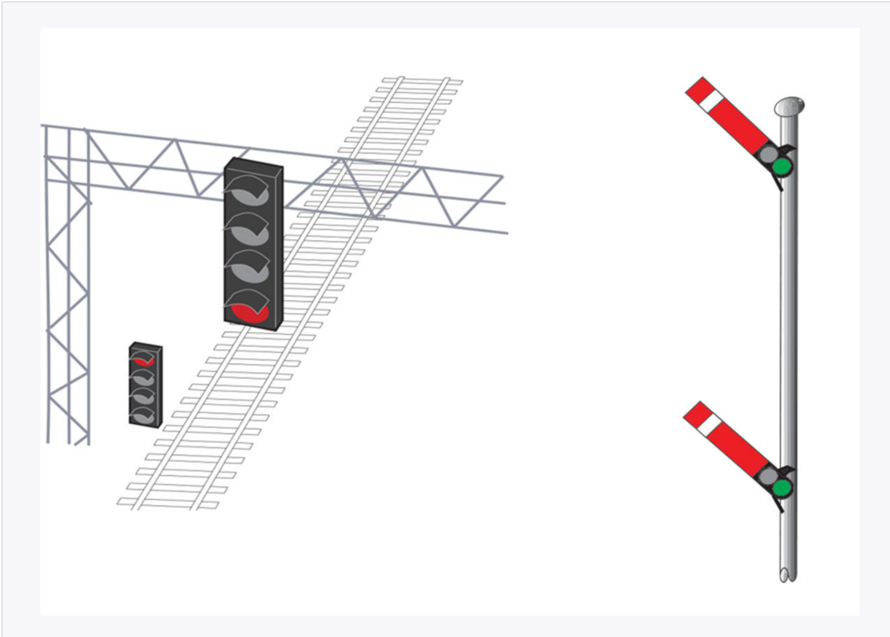


3-state banner repeater showing the green-lit OFF indication — RS521 §5.7

8.3 Co-Acting Signals (CA)

Co-acting signals are provided to give both short and long distance sighting of the same signal. They always:

- Repeat the EXACT aspect or indication of the main signal.
- Are always the same type (colour light or semaphore) as the main signal.
- Are identified by "CA" on the signal identification plate.



Co-acting signals give both distant and close-range sighting of the same aspect — RS521 §5.7

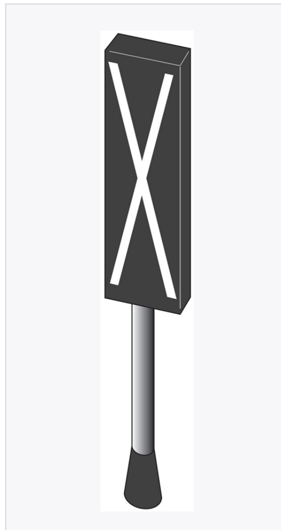
9. Signals Not in Use

When a colour light signal is not in use:

- Main and position-light signals are covered up.
- Main aspects may also have a large "X" displayed over the cover.

When a semaphore signal is not in use:

A large X is fixed on the signal arm, or the disc is covered over.



A signal covered and marked with a large X — RS521

10. Key Rules to Remember

1. Red is ALWAYS at the BOTTOM of a colour light signal.
2. Always reduce speed at any caution signal.
3. Automatic signals will not usually have TPWS.
4. Travelling between different signalling systems creates risk.
5. When a signal is not showing: stop signal = treat as danger; distant = treat as caution.
6. On position-light signals: always be prepared to stop short of any obstruction, buffer stop, train or signal at stop.
7. Wait for the main aspect before acting — never proceed on a route indicator or junction indicator alone.

Questions and Answers

Q1. What are the four components of a track circuit?

A: Power source, insulated rail joints (insulated gaps), track circuit relay, and the rails (track circuit itself). A train's axles short-circuit the rails, dropping the relay and holding the signal at danger.

Q2. How do axle counters work?

A: They count axles as a train enters and exits a section, confirming the complete train has cleared before the section is considered free.

Q3. What is the overlap distance in colour light signalling?

A: 200 metres beyond the signal — the area kept clear in case a train overruns a stop signal.

Q4. What is the clearance point distance in absolute block signalling?

A: 400 metres beyond the signal — the distance a train must have passed before the section ahead is considered clear.

Q5. How does a controlled signal identification plate differ from an automatic signal plate?

A: A controlled signal has a plain black plate with white text. An automatic signal has a black plate with a white horizontal stripe across the top.

Q6. What does 'SEMI' on a signal identification plate mean?

A: It identifies a semi-automatic signal — normally operated by the passage of trains but can also be controlled from a signal box or ground frame.

Q7. What does 'R' on a signal identification plate indicate?

A: A distant signal (cannot show a stop aspect). 'RR' indicates an outer distant signal.

Q8. What do the letters 'BR' and 'CA' on signal plates mean?

A: 'BR' = banner repeating signal (provided at signals with restricted sighting). 'CA' = co-acting signal (mirrors the main signal to aid sighting).

Q9. In three-aspect signalling, what does a single yellow aspect mean?

A: Caution — be prepared to stop at the next signal.

Q10. In four-aspect signalling, what does a double yellow aspect mean?

A: Preliminary caution — be prepared to find the next signal at single yellow.

Q11. What is the correct action if a colour light stop signal is not showing any aspect?

A: Treat it as being at danger (GERT8000-S7 §1.5).

Q12. What does a flashing yellow aspect mean?

A: Facing points at a junction ahead are set for a diverging route with a lower speed than the straight route. The route IS set — it is advance information about the junction, not a general caution.

Q13. How many preliminary route indicators are provided in four-aspect signalling, and how many in three-aspect?

A: Four-aspect: 3 preliminary route indicators. Three-aspect: 2 preliminary route indicators.

Q14. What does a preliminary route indicator show, and when is it not illuminated?

A: It displays an arrow pointing in the same direction as the junction indicator at the signal ahead. It is not illuminated when the junction signal is at danger.

Q15. What does a position-light signal showing two white lights at 45° mean?

A: Proceed at caution towards the next train, signal or buffer stop — be prepared to stop short of any obstruction.

Q16. What is the normal state of a position-light signal associated with a main aspect?

A: Unlit — meaning obey the main signal.

Q17. Can a driver proceed with a passenger train on the authority of a position-light signal alone?

A: No — not unless entering a permissive platform line on the authority of a position-light or subsidiary signal associated with a main aspect (GERT8000-S7 §3.1).

Q18. What must a driver do when the main aspect is red but a position-light signal shows proceed (permissive working)?

A: Drive at caution, be prepared to stop short of any obstruction including another train, vehicle or buffer stop.

Q19. What is a banner repeating signal and when is it used?

A: It gives advance information about the aspect of a signal ahead that has restricted sighting (e.g. around a curve, in a tunnel, behind buildings). Position OFF = signal showing proceed; position ON = signal at danger/caution.

Q20. What is the difference between a 2-state and 3-state banner repeating signal?

A: A 2-state banner shows ON or OFF — OFF means the signal is showing a proceed aspect (yellow or green). A 3-state banner adds a specific green OFF indication, confirming the signal ahead is showing green specifically.

Q21. What does a co-acting signal do and how is it identified?

A: It repeats the exact aspect of the main signal to aid sighting from close range. It is always the same type as the main signal and is identified by 'CA' on its plate.

Q22. What is the risk when travelling between different signalling systems?

A: The transition creates risk as the driver must adapt to a different signal sequence and rules. In particular, on ERTMS-operated trains, the ability for signals to display flashing aspects is disabled — only standard aspects are shown.

Q23. What is a splitting distant signal and what does it show?

A: A splitting distant signal is provided at diverging junctions and has a primary head and an off-set head. It shows which route is set at the junction ahead — the lit heads indicate what aspect is shown at the first signal after the junction for that route.

Q24. When a signal shows a proceed aspect with a junction indicator, when should the driver proceed?

A: Only when the MAIN ASPECT shows proceed — never on the junction indicator alone. Wait until the main aspect is confirmed before moving.

Q25. What should a driver do if stopped before the whole train has passed a signal that is showing proceed?

A: The driver may act on the aspect that was being displayed when they passed the signal, unless instructed not to proceed (GERT8000-S7 §2.2).